

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import LabelEncoder
```

```
In [18]: df=pd.read_csv("Social_Network_Ads.csv")
```

```
In [19]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   User ID               400 non-null   int64
1   Gender                400 non-null   object
2   Age                  400 non-null   float64
3   EstimatedSalary       400 non-null   float64
4   Purchased             400 non-null   int64
dtypes: float64(2), int64(2), object(1)
memory usage: 15.8+ KB
```

```
In [41]: df
```

```
Out[41]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	1	19.0	19000.0	0
1	15810944	1	35.0	20000.0	0
2	15668575	0	26.0	43000.0	0
3	15603246	0	27.0	57000.0	0
4	15804002	1	19.0	76000.0	0
...	...	...	...	...	...
395	15691863	0	46.0	41000.0	1
396	15706071	1	51.0	23000.0	1
397	15654296	0	50.0	20000.0	1
398	15755018	1	36.0	33000.0	0
399	15594041	0	49.0	36000.0	1

400 rows × 5 columns

```
In [21]: df.isnull().sum()
```

```
Out[21]: User ID          0
Gender          0
Age            0
EstimatedSalary 0
Purchased       0
dtype: int64
```

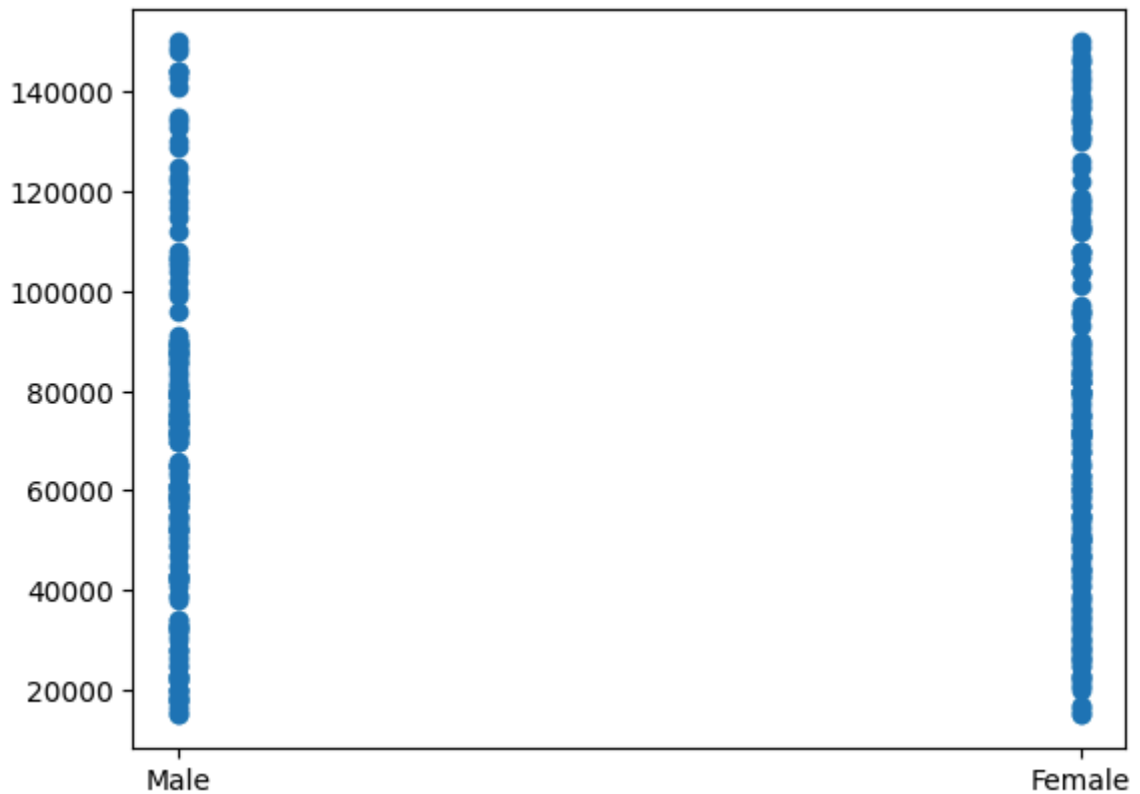
```
In [40]: df['Purchased']
```

```
Out[40]: 0      0
          1      0
          2      0
          3      0
          4      0
          ..
        395    1
        396    1
        397    1
        398    0
        399    1
        Name: Purchased, Length: 400, dtype: int64
```

```
In [16]: df.isnull().sum()
```

```
Out[16]: CRIM      20
          ZN      20
          INDUS    20
          CHAS    20
          NOX      0
          RM      0
          AGE     20
          DIS      0
          RAD      0
          TAX      0
          PTRATIO  0
          B        0
          LSTAT    20
          MEDV     0
          dtype: int64
```

```
In [25]: plt.scatter(df['Gender'],df['EstimatedSalary'])
          plt.show()
```



```
In [28]: from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
df['Gender']=le.fit_transform(df['Gender'])
```

```
In [ ]:
```

```
In [32]: from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

```
In [29]: df.columns
```

```
Out[29]: Index(['User ID', 'Gender', 'Age', 'EstimatedSalary', 'Purchased'], dtype='object')
```

```
In [33]: x=df[['Gender', 'Age', 'EstimatedSalary']]
y=df[ 'Purchased']
```

```
In [34]: x_train,x_test,y_train,y_test =train_test_split(x,y,test_size=0.2,random_state=0)
```

```
In [48]: from sklearn.linear_model import LogisticRegression
# instantiate the model (using the default parameters)
logreg = LogisticRegression(max_iter=400)
logreg.fit(x_train, y_train)
```

```
Out[48]: LogisticRegression
LogisticRegression(max_iter=400)
```

```
In [49]: y_pred = logreg.predict(x_test)
print(y_pred)
```

```
[0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 0 0 0 1 0 0 1 0 0 0 1 0 0 0 0 1 0 0 0
 0 0 1 0 1 1]
```

```
In [50]: from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [51]: print(f"accuracy={accuracy_score(y_test,y_pred)}")
```

accuracy=0.925

```
In [53]: print(f"confusion matrix=\n{confusion_matrix(y_test,y_pred)}")
```

confusion matrix=

```
[[56  2]
 [ 4 18]]
```

```
In [54]: df.groupby('Gender')['Purchased'].sum()
```

```
Out[54]: Gender
0      77
1      66
Name: Purchased, dtype: int64
```

```
In [ ]:
```